

Spatially Explicit Rock–TBM Interaction Graph Sequences for Response-Consistent Interpretation of TBM Excavation Processes

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Abstract

Existing representations of rock–TBM interaction during excavation treat monitoring responses as temporal sequences, which preserve temporal dependence but lack the spatial explicitness needed to resolve where interactions occur, which machine components are implicated, and how interaction structure evolves along chainage. This study proposes a response-supervised, geometry-constrained graph sequence framework for spatially explicit representation and interpretable learning of rock–TBM interaction. The framework introduces three key innovations: (i) a heterogeneous graph-sequence representation that formalizes TSP-derived rock voxels and parameterized TBM surface nodes as spatial entities with explicit candidate interaction relations; (ii) a geometry-constrained edge construction strategy based on spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints, which restricts learning to spatially plausible rock–machine relations rather than unconstrained connectivity; and (iii) a graph-to-surface interpretation route that projects learned edge relevance onto TBM surface and chainage spaces to produce response-consistent hotspot maps and interaction-evolution views. Rather than relying on sparse jamming labels or manually calibrated contact forces, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations, enabling interpretable spatial learning under incomplete observability. The evaluation uses two tunnel cases: BSLL DyK1017+205 and SJLS Dyk1252+411. BSLL is used for one-step, multi-step, and interpretation analysis, while SJLS provides an external TSP case for testing prediction consistency and geometry ablation. The framework remains competitive with temporal baselines, and SJLS shows a small geometry-sensitive MAE advantage, but the experiments do not establish a broad statistically significant forecasting improvement. The stronger evidence is interpretive: checkpoint-based diagnostics show positively clustered surface relevance in all evaluated runs, relevance that is not simply explained by candidate-edge degree, and case-specific association with local TSP velocity and monitoring responses. These patterns are reported as response-consistent diagnostics, not as proof of physical contact or geological causality. By unifying geological sensing, machine geometry, monitoring response, and spatial interpretation within a constrained graph sequence, the framework offers a spatially explicit approach to excavation interaction analysis that complements existing temporal prediction methods.

Keywords: spatial representation; heterogeneous graph sequence; rock–machine interaction; geometry-constrained edge construction; response supervision; interpretable geospatial learning; TBM excavation; spatial interpretation

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1 Introduction

Tunnel boring machines (TBMs) are widely used in long and deep tunnel construction, where excavation responses emerge from the coupled effects of geological conditions, machine geometry, and operational process. Under fractured, water-rich, high-stress, or locally deformable ground conditions, the interaction state between surrounding rock and the TBM may change rapidly and manifest through abnormal thrust, torque, penetration, advance rate, or shield-related responses. These monitoring variables are informative external expressions of rock–TBM interaction, but monitoring curves alone provide limited evidence about where the interaction occurs, which machine components are implicated, and how local interaction relations evolve along chainage.

Recent developments in tunnel seismic prediction (TSP), advance geological prediction, digital tunnelling, and TBM monitoring systems have substantially increased the availability of heterogeneous excavation data [11]. TSP provides a spatially distributed description of the geological field ahead of or around excavation, whereas monitoring systems provide temporally continuous responses referenced to excavation progress. The challenge is therefore not only data acquisition, but also data organization: heterogeneous sources with different spatial support, temporal resolution, and physical meaning must be integrated into a representation that preserves excavation evolution together with local spatial structure.

This challenge can be viewed as a geographic information representation problem. Three GIScience concerns are central. First, *spatial entity formalisation*: TSP voxels, TBM surface components, and monitoring locations are heterogeneous spatial objects with distinct spatial supports (point, surface, volume) and semantic roles, and must be abstracted into a unified entity model that preserves their geometric and topological properties [4, 6, 9]. Second, *constrained relation construction*: candidate rock–machine relations cannot be learned freely from black-box correlation but must respect geometric compatibility, excavation state, and active-zone constraints—an instance of the broader GIScience principle that spatial relations should be constructed with domain-plausible topology rather than unconstrained connectivity [3, 14, 29]. Third, *interpretable spatial learning*: the learned representation must be projectable back to interpretable spatial views (surface hotspots, chainage evolution) to support spatial reasoning, not merely optimise a prediction loss [10, 13]. In this light, the problem extends beyond conventional engineering prediction and can be framed as a GIScience problem—one that benefits from explicit spatial entity modelling, constrained relation construction, and interpretable spatial learning rather than merely feature concatenation or temporal sequence extension.

Existing TBM prediction and warning studies have shown that tabular models and sequence models can be effective for forecasting penetration, thrust, torque, or jamming-related responses [1, 7, 24, 25]. However, most of these methods represent excavation either as a feature table or as an ordered monitoring sequence. Sequence models are effective for learning temporal dependence in monitoring records [8, 18], but they usually treat excavation as an ordered signal sequence rather than as a spatially structured interaction process. As a consequence, they have limited ability to preserve rock location, machine-component structure, and local rock–machine candidate relations within a unified representation. This limitation matters when the objective is not only response prediction, but also spatially consistent interpretation of interaction dynamics.

Graph learning offers a natural alternative because it is well suited to irregular objects and relations. Graph-based models have shown strong capacity for encoding object interaction, state propagation, and structured spatial systems in domains such as physics simulation [15, 16], urban flow prediction [5, 27], and spatio-temporal forecasting [12, 23]. Heterogeneous graph networks further enable the representation of multi-type entities and relations [17, 21, 26]. Yet TBM excavation is not a generic graph-learning problem in which edges can be learned freely from black-box correlation. Candidate rock–TBM relations must remain consistent with machine geometry, excavation state, and local active zones. In this setting, graph learning is

most meaningful when it operates on geometrically plausible candidate relations rather than on unconstrained connectivity.

Motivated by this gap, this study proposes a response-supervised, geometry-constrained graph sequence framework for spatially explicit rock–TBM interaction representation and interpretation. TSP-derived geological attributes are organized as rock voxel nodes, while the cutterhead and shield are discretized as parameterized TBM surface nodes. Candidate rock–machine edges are screened through spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints. Instead of interpreting edge weights as direct physical contact forces or relying on sparse jamming labels, the framework uses continuously recorded monitoring responses to supervise the relative importance of geometry-screened candidate relations. The learned relevance is then projected back to TBM surface space and chainage space to generate response-consistent hotspot and evolution views.

The study addresses three questions: how to organize TSP rock attributes, TBM surface structure, and rock–machine candidate relations into a dynamic graph sequence that preserves spatial explicitness and relation plausibility; how to learn the relative importance of candidate relations from continuous monitoring responses without sparse event labels, enabling interpretable spatial learning under incomplete observability; and how to map learned interaction relevance back to interpretable spatial views that support spatial reasoning beyond prediction. The main contributions are fourfold.

1. *A spatially explicit heterogeneous graph-sequence representation for rock–machine interaction.* Unlike feature-table or pure-sequence representations that discard spatial structure, the proposed representation formalises TSP voxels, TBM surface components, and their candidate relations as typed spatial entities with explicit geometric and topological properties, preserving spatial explicitness throughout the learning pipeline.
2. *A geometry-constrained relation construction strategy that embeds domain plausibility into graph topology.* Candidate rock–machine edges are screened by spatial proximity, signed-normal consistency, active-zone filtering, and excavation-state constraints. This ensures that the learned relations remain geometrically plausible, addressing the GIScience principle that spatial relations should be constructed with domain-plausible topology rather than unconstrained connectivity.
3. *A response-supervised relation learning paradigm that operates under incomplete observability.* Without requiring contact-force labels or sparse jamming events, the framework uses continuously recorded monitoring responses to supervise the relative importance of candidate relations, enabling interpretable spatial learning in settings where ground-truth interaction labels are unavailable.
4. *A graph-to-surface interpretation route that supports spatial reasoning beyond prediction.* Learned interaction relevance is projected back to TBM surface hotspots and chainage-evolution views, accompanied by spatial consistency indicators (Moran’s I , component CV, attention–geology correlation) that quantify the spatial plausibility of the learned representation.

2 Methodology

This section formalizes the proposed framework as a geometry-constrained heterogeneous graph sequence supervised by continuously recorded TBM monitoring response. The method represents excavation as an evolving system of spatial entities—rock units and TBM surface components—and their candidate interaction relations, rather than as a monitoring-only signal sequence. By formalizing spatial entities, constraining candidate relations by geometric compatibility, and projecting learned relevance back to interpretable spatial views, the framework

addresses the geographic-information representation problem identified in the Introduction: how to organize heterogeneous spatial entities into a representation that preserves spatial explicitness, relation plausibility, and interpretability.

2.1 Framework overview

Let excavation advance be discretized into chainage-indexed steps $t = 1, 2, \dots, T$. For each step, the framework constructs a graph snapshot G_t and then models the ordered graph sequence $\{G_t\}_{t=1}^T$. The workflow contains three stages: geometry-constrained graph-sequence construction, response-supervised interaction learning, and graph-to-surface spatial interpretation (Figure 1). Given the most recent K excavation steps, the model uses graph snapshots together with synchronized TBM monitoring vectors to predict a future response at step $t + h$:

$$\hat{r}_{t+h} = f_{\Theta}(G_{t-K+1:t}, u_{t-K+1:t}),$$

where u_t denotes the monitoring vector at step t , r_{t+h} denotes the target future response, K is the historical window length, and h is the forecasting horizon. In this study, the learned rock-machine edge attention is interpreted as response-consistent interaction relevance under the forecasting task, not as true contact force, contact pressure, jamming probability, or any direct engineering risk quantity.

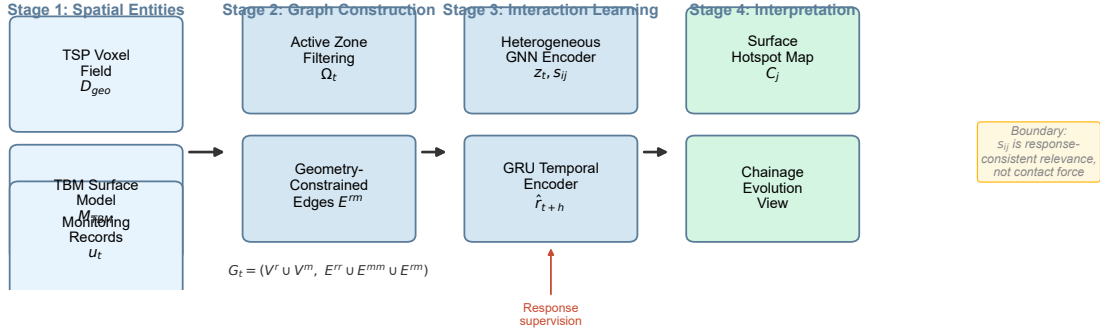


Figure 1: Framework pipeline (schematic). Node counts and edge counts are downsampled for visual clarity; the actual graph uses 1352 TBM surface nodes and approximately 3500 active-zone rock nodes per step (see Implementation details). Stage 1 constructs spatial entities from the TSP voxel field, TBM surface geometry, and monitoring records. Stage 2 builds geometry-constrained graph snapshots through active-zone filtering and candidate rock-machine edge construction. Stage 3 performs response-supervised graph-sequence learning using a heterogeneous GNN encoder and GRU temporal encoder. Stage 4 projects learned relevance back to TBM surface and chainage spaces to generate hotspot maps and evolution views.

2.2 Excavation-step definition and data alignment

Raw TBM monitoring records are usually sampled at a much higher frequency than the graph sequence should use. The method therefore first aggregates monitoring data by chainage or ring-based excavation progress to form a unified excavation-step index t . A practical initial aggregation granularity is 0.05 m or 0.1 m, subject to data density and desired spatial resolution. Each excavation step is associated with four objects: current face or shield position, graph snapshot G_t , synchronized monitoring vector u_t , and future response label r_{t+h} .

The monitoring input vector can include

$$u_t = [\text{AdvanceRate}_t, \text{RPM}_t, \text{Torque}_t, \text{Thrust}_t, \text{Penetration}_t, \text{ShieldPressure}_t],$$

while the prediction target can be defined as a multivariate future response vector

$$r_{t+h} = [\text{AdvanceRate}_{t+h}, \text{Torque}_{t+h}, \text{Thrust}_{t+h}, \text{Penetration}_{t+h}, \text{ShieldPressure}_{t+h}].$$

Because TSP coordinates and TBM shield mileage are usually recorded in different local reference systems, the method also requires a local offset mapping. A simple alignment is to define

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

and then map $x_{\text{local}}(t)$ into the TSP coordinate range used for voxelized geological representation.

2.3 Geological and machine inputs

The geological input is a TSP-derived voxel field $D_{\text{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ denotes the spatial coordinate of rock voxel i , and g_i denotes its geological attribute vector. Depending on the available inversion and interpretation products, g_i may include seismic velocity, elastic indicators, ratio features, or other derived geological descriptors.

The TBM input is represented as a parameterized surface model $M_{\text{TBM}} = \{p_j(0), n_j, \rho_j\}$, where $p_j(0)$ is the initial coordinate of TBM surface node j , n_j is the local surface normal, and ρ_j is a machine component label such as cutterhead or shield. The excavation process is further described by advance direction d , step length Δx , step-dependent active zone Ω_t , interaction distance threshold τ , and normal-compatibility threshold $\eta_{\min}$. When a detailed three-dimensional TBM model is unavailable, the surface can be approximated by a parameterized geometry in which the cutterhead is represented as a disc and the shield as cylindrical surface components. In the current formulation, the surface is partitioned into cutterhead, front shield, middle shield, and tail shield, and nodes are sampled at fixed radial, circumferential, and axial resolutions.

2.4 Graph-sequence construction

At each excavation step t , a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}),$$

where V_t^r and V_t^m are rock and TBM node sets, respectively, E_t^{rr} denotes rock–rock adjacency, E_t^{mm} denotes TBM-surface adjacency, and E_t^{rm} denotes candidate rock–machine interaction edges.

Rock node features are defined as

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether a voxel remains active and eligible for interaction at step t . TBM node features are defined as

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

With constant advance step length, TBM surface nodes are updated along the advance direction by

$$p_j(t+1) = p_j(t) + \Delta x d.$$

For intra-rock connectivity, E_t^{rr} is constructed from voxel adjacency, for example a 26-neighbourhood relation. For TBM-surface connectivity, E_t^{mm} is constructed from local surface neighbourhood relations inherited from the discretized machine geometry. The key design choice concerns the candidate interaction edges E_t^{rm} , which are not fully connected but screened by geometric plausibility. This constrained construction contrasts with unconstrained graph learning [22, 28] and reflects the domain requirement that candidate rock–machine relations must be physically plausible.

2.5 Geometry-constrained candidate interaction edges

A rock voxel i and TBM surface node j are linked at step t only if the candidate relation satisfies a geometry-constrained rule:

$$e_{ij}^{rm}(t) = 1$$

if

$$\begin{cases} d_{ij}(t) \leq \tau, \\ \kappa_{ij}(t) \geq \eta_{\min}, \\ c_i \in \Omega_t, \\ s_i(t) = 1, \end{cases}$$

and the edge is absent otherwise. Here,

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

is the rock-to-surface distance, and

$$\kappa_{ij}(t) = \max \left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon} \right)$$

measures signed-normal compatibility between the TBM surface orientation and the rock-voxel direction. The active-zone condition restricts candidate relations to the geological region currently relevant to excavation, while the state variable $s_i(t)$ prevents already excavated or inactive voxels from repeatedly contributing to later interaction snapshots.

This design intentionally limits graph learning to a screened candidate set. The edge relevance learned later in the model is therefore interpreted as the relative importance of geometrically plausible candidate interactions, rather than as a direct estimate of physical contact force or stress.

2.6 Edge features and geometric prior

Each candidate rock-machine edge is further associated with an edge-feature vector

$$a_{ij}^{rm} = [d_{ij}/\tau, \kappa_{ij}, (c_i - p_j)/\tau, g_i, \text{onehot}(\rho_j)],$$

which combines three kinds of information: geometric compatibility, local geological context, and machine-component semantics. In addition, a simple geometric prior is defined as

$$\pi_{ij}^{rm} = \exp(-d_{ij}/\tau) \kappa_{ij}.$$

This prior increases when the rock voxel is closer to the TBM surface and more consistent with the local outward normal. It is used only as geometric guidance for learning and should not be interpreted as contact force, contact pressure, or a direct risk indicator. The full graph-sequence construction procedure is summarised in Algorithm 1.

2.7 Response-supervised interaction learning

After graph construction, each snapshot is encoded by a graph encoder to produce an interaction-aware latent representation. Message passing integrates geological attributes, machine-surface context, and candidate interaction structure across the heterogeneous graph. The resulting stepwise graph embeddings are then passed to a temporal encoder, which models graph-state evolution along chainage and links the sequence of graph states to continuously recorded TBM monitoring response.

For each graph snapshot, the graph encoder computes raw (pre-softmax) edge scores $s_{ij}^{rm}(t)$ for each candidate rock–machine edge, which are then normalised over all source rock nodes incident to each destination TBM node j to obtain attention weights $\alpha_{ij}^{rm}(t) = \exp(s_{ij}^{rm}(t)) / \sum_{i'} \exp(s_{i'j}^{rm}(t))$ used in message passing [19, 20]. The encoder also produces a graph-level representation z_t . The graph-level representation is concatenated with the synchronized monitoring vector u_t and passed to a gated recurrent unit (GRU) [2], which models graph-sequence evolution across consecutive excavation steps.

The supervision signal is provided by monitoring response variables recorded during excavation, rather than by sparse binary jamming labels or manually assigned edge weights. Under this design, the model learns which screened candidate rock–machine relations are most relevant for explaining or anticipating response variation. The resulting raw scores $s_{ij}^{rm}(t)$ are therefore response-consistent relevance indicators under geometric screening. The training procedure is summarised in Algorithm 2.

2.8 Graph-to-surface spatial interpretation

To recover spatial interpretability, the learned relevance associated with candidate rock–machine relations is projected back onto two explicit spaces.

Notation. We distinguish three quantities in the interpretation pipeline:

- *Raw edge score* $s_{ij}^{rm}(t)$: the pre-softmax attention score computed by the graph encoder for candidate edge (i, j) at step t . This quantity preserves absolute magnitude differences across edges and is used for all spatial interpretation outputs.
- *Normalised attention weight* $\alpha_{ij}^{rm}(t)$: the post-softmax weight used in message passing, obtained by normalising raw scores over all source rock nodes incident to destination TBM node j :

$$\alpha_{ij}^{rm}(t) = \frac{\exp(s_{ij}^{rm}(t))}{\sum_{i': (i', j) \in E_t^{rm}} \exp(s_{i'j}^{rm}(t))}.$$

Because $\sum_{i: (i, j) \in E_t^{rm}} \alpha_{ij}^{rm}(t) = 1$ for each TBM node j , aggregating α_{ij}^{rm} per node yields a constant and is therefore unsuitable for spatial interpretation.

- *Surface relevance* $C_j(t)$: the degree-normalised aggregated non-negative-transformed raw score incident to TBM surface node j , defined as

$$C_j(t) = \frac{1}{d_j(t) + \epsilon} \sum_{i: (i, j) \in E_t^{rm}} \text{softplus}(s_{ij}^{rm}(t)),$$

where $d_j(t) = |\{i : (i, j) \in E_t^{rm}\}|$ is the degree of TBM node j in the rock–machine sub-graph at step t , $\epsilon = 10^{-8}$ avoids division by zero, and $\text{softplus}(x) = \log(1 + e^x)$ is a smooth non-negative transform. The softplus is introduced because raw pre-softmax scores s_{ij}^{rm} can be negative, and direct summation would allow positive and negative scores to cancel, producing ambiguous C_j values that confound spatial interpretation. Softplus preserves the monotonic ordering of raw scores while ensuring non-negativity, so that C_j faithfully reflects the aggregate interaction relevance incident to each surface node. The degree normalisation prevents nodes with more incident candidate edges from receiving artificially inflated relevance solely due to edge count, ensuring that C_j reflects the *average per-edge relevance* rather than the raw sum. No further per-chainage standardisation (e.g. z-scoring) is applied to C_j , so that absolute magnitude differences across chainages are preserved for downstream spatial consistency indicators (Moran’s I , component CV, and attention–geology correlation all require non-degenerate per-chainage means and variances). For hotspot visualisation only, C_j is min–max rescaled per chainage for colour

mapping; the underlying values used in all quantitative indicators remain the unstandardised degree-normalised scores.

Projection. First, $C_j(t)$ is projected onto the cutterhead and shield surface to produce component-aware hotspot maps. Second, the sequence of $C_j(t)$ values is tracked along excavation chainage to visualize how interaction emphasis evolves over space and excavation progress. These outputs show where the model places interaction importance and how that importance migrates during excavation, while remaining consistent with the monitored response signal used for supervision. The interpretation procedure is summarised in Algorithm 3.

2.9 Implementation details

The framework is implemented in Python using PyTorch and PyTorch Geometric. Key hyperparameters and structural settings are summarised below. The same graph-construction settings are used in the Experimental setup (Section 3.5) and all ablations; the prediction horizon is set by each case configuration.

Graph structure. The TBM surface is discretised into 1352 nodes (cutterhead disc + front, middle, and tail shield cylinders) at a surface resolution of 0.5 m. The TSP voxel field yields approximately 3500 active-zone rock nodes per excavation step (filtered by the component-specific active zone: cutterhead look-ahead 5.0 m with radial extent $\tau_{\text{zone}} = 5.0$ m; shield annular ring with $R_s \leq r \leq R_s + \tau_{\text{zone}}$). Candidate rock-machine edges are constructed with distance threshold $\tau_{\text{edge}} = 2.0$ m and signed-normal compatibility threshold $\eta_{\text{min}} = 0.3$. The historical window length is $K = 5$. The BSLL one-step and SJLS cases use $h = 1$, while the BSLL multi-step sensitivity case uses $h = 3$.

Model architecture. The heterogeneous GNN encoder uses $\text{gnn_layers} = 2$ message-passing layers with rock-node hidden dimension 32, TBM-node hidden dimension 32, and edge hidden dimension 32; dropout is 0.3. The GRU temporal encoder has a hidden dimension of 64 with 1 layer. The prediction head is a two-layer MLP with ReLU activation. Baseline LSTM and TSP-LSTM models use hidden dimension 128, 2 layers, and dropout 0.1 for comparability.

Training. All models are trained for up to 80 epochs using AdamW optimiser with initial learning rate 10^{-3} , weight decay 10^{-2} , standardised Huber loss ($\delta = 1.0$), and ReduceLROnPlateau scheduler (factor 0.5, patience 10). Batch size is 32. Early stopping with patience 15 is applied on validation loss. Monitoring variables and TSP attributes are standardised using training-set statistics only. Monitoring scalars are fitted from rows used by the chronological training samples, and TSP attribute scalars are fitted from voxels in the neighbourhood of training target chainages. Random seed is fixed at 42 for reproducibility.

Spatial consistency indicators. Moran’s I is computed using inverse-distance weights on TBM surface node positions within each chainage. Component CV is the coefficient of variation of mean C_j across the four TBM components (cutterhead, front, middle, tail shield). Attention-geology correlation is the Pearson and Spearman correlation between mean C_j and TSP V_p across test chainages.

2.10 Evaluation design

The experimental design compares the proposed framework against several baseline families: monitoring-only sequence models, TSP-augmented sequence models, tabular baselines, and graph-structure ablations. Structural ablations include randomized rock-machine edges and versions without geometric constraints. This design isolates the contribution of graph structure, geological attributes, temporal graph evolution, and geometry-based candidate screening.

The formal evaluation uses a chronological mileage-ordered split. The first 70% of prediction-target samples are used for training, the next 15% for validation, and the remaining 15% for

testing. This design evaluates forward generalisation to a later continuous tunnel interval and deliberately retains the geological distribution shift encountered by the advancing TBM.

2.11 Algorithmic summary

The three stages of the framework are summarised in Algorithms 1–3.

Algorithm 1 Geometry-constrained graph-sequence construction

Require: Rock voxel field D_{geo} , TBM surface M_{TBM} , thresholds $\tau, \eta_{\min}$, active-zone Ω_t

- 1: **for** each excavation step $t = 1, \dots, T$ **do**
- 2: Update TBM surface positions: $p_j(t) \leftarrow p_j(t-1) + \Delta x d$
- 3: Build rock–rock edges E_t^{rr} from 26-neighbourhood
- 4: Build TBM–TBM edges E_t^{mm} from surface adjacency
- 5: Build candidate rock–machine edges:

$$E_t^{rm} \leftarrow \{(i, j) : d_{ij}(t) \leq \tau, \kappa_{ij}(t) \geq \eta_{\min}, c_i \in \Omega_t, s_i(t) = 1\}$$

- 6: Compute edge features a_{ij}^{rm} and geometric prior π_{ij}^{rm}
 - 7: Assemble graph snapshot $G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$
 - 8: **end for**
 - 9: **return** Graph sequence $\{G_t\}_{t=1}^T$
-

Algorithm 2 Response-supervised interaction learning

Require: Graph sequence $\{G_t\}$, monitoring vectors $\{u_t\}$, window K , horizon h

- 1: **for** each training step **do**
 - 2: **for** each sample window $t - K + 1$ to t **do**
 - 3: Encode each G_s by heterogeneous GNN: $z_s, \{s_{ij}^{rm}(s)\} \leftarrow \text{GNN}(G_s)$
 - 4: Concatenate graph and monitoring: $h_s = [z_s; u_s]$
 - 5: **end for**
 - 6: Model temporal evolution: $\{h_s\} \leftarrow \text{GRU}(\{h_s\})$
 - 7: Predict: $\hat{r}_{t+h} = \text{FC}(h_T)$
 - 8: Compute loss: $\mathcal{L} = \text{Huber}(\hat{r}_{t+h}, r_{t+h})$
 - 9: Update parameters Θ by back-propagation
 - 10: **end for**
 - 11: **return** Trained model f_{Θ} , raw edge scores $\{s_{ij}^{rm}(t)\}$
-

Algorithm 3 Graph-to-surface spatial interpretation

Require: Trained model f_Θ , test graph sequence, TBM surface

- 1: **for** each test chainage c **do**
 - 2: Extract pre-softmax edge scores $s_{ij}^{rm}(c)$ for rock-machine edges
 - 3: Aggregate per TBM node (degree-normalised): $C_j(c) = \frac{1}{d_j(c)+\epsilon} \sum_{i:(i,j) \in E^{rm}} \text{softplus}(s_{ij}^{rm}(c))$
 - 4: Project $C_j(c)$ onto TBM surface $\rightarrow$ hotspot map at chainage c
 - 5: **end for**
 - 6: Track $C_j(c)$ along chainage $\rightarrow$ component-wise evolution view
 - 7: **Output:** Hotspot maps, chainage-evolution view
-

3 Experiments and results

3.1 Case study data

The experiments use two real tunnel cases. The first is the BSLL tunnel case starting from DyK1017+205. It provides the main graph-sequence construction example, the one-step response prediction run, and the three-step advance-prediction sensitivity run. The second is the SJLS tunnel case starting from Dyk1252+411. It uses an external TSP-derived velocity field and real mileage-aligned TBM monitoring labels, and is used to test whether the same spatial representation remains meaningful in a second tunnel setting.

The two cases are not treated as sequential data releases or replacement datasets. They are two case-study settings with different evidential roles. BSLL is used mainly to illustrate multi-step prediction and spatial interpretation under a compact test interval. SJLS is used mainly for prediction consistency and geometry-ablation evidence under an external TSP case.

3.2 Monitoring and geological variables

For each excavation step, the monitoring vector contains advance rate, cutterhead RPM, torque, thrust, penetration, and shield pressure. The prediction target is the future multivariate response vector consisting of advance rate, torque, thrust, penetration, and shield pressure. These variables describe the observed machine response to the surrounding rock and are used both as historical inputs and as response-supervision targets.

The geological input is a TSP-derived voxel field. For BSLL, the project TSP voxel field contains P-wave velocity, S-wave velocity, density, dynamic Young’s modulus, velocity ratio, and Poisson’s ratio around the tunnel alignment. For SJLS, dense external P-wave and S-wave velocity profiles at Dyk1252+411 are preprocessed into the same project TSP schema and paired with mileage-aligned monitoring labels. This preprocessing allows both tunnel cases to use the same graph-construction and graph-sequence learning pipeline.

3.3 Mileage-ordered partitioning

All experiments use mileage-ordered partitioning to preserve forward prediction. For a historical window of $K = 5$, each sample uses the preceding monitoring and graph snapshots as input and predicts a future response at horizon h . The formal case runs are summarised in Table 1. The BSLL $h=1$ run evaluates one-step prediction, the BSLL $h=3$ run evaluates multi-step advance prediction, and the SJLS run evaluates the external TSP case.

3.4 Scope and interpretive boundaries

The experiments evaluate a spatial representation rather than a universally superior forecasting model. Three case runs are used: BSLL DyK1017+205 with one-step prediction, BSLL

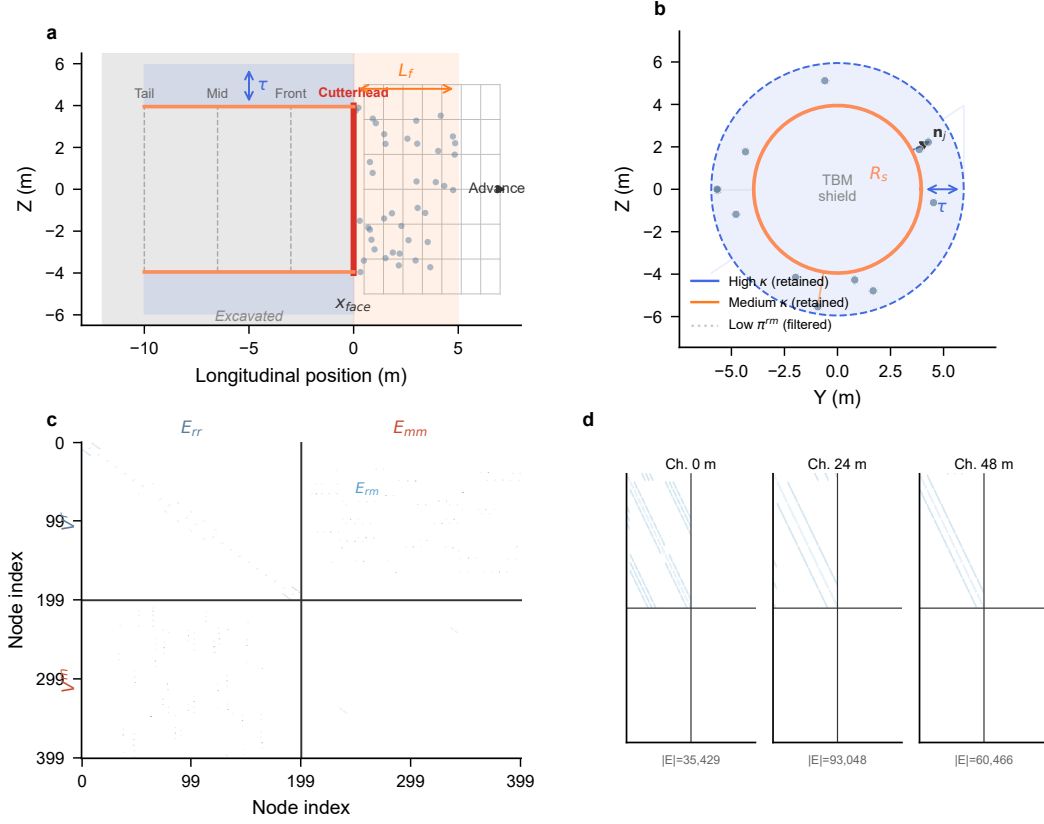


Figure 2: Schematic of geometry-constrained graph construction at a representative excavation step (illustrative, not drawn from the actual TSP velocity field). The framework links TSP-derived rock voxels, parameterized TBM surface nodes, and geometry-screened rock-machine candidate edges before learning response-consistent relevance.

Table 1: Formal case runs used in the experiments. Effective samples are sequence windows after applying the history window and prediction horizon.

Case	Tunnel	Start chainage	Horizon	Train/Test samples
BSLL h=1	BSLL	DyK1017+205	1	30 / 8
BSLL h=3	BSLL	DyK1017+205	3	29 / 7
SJLS h=1	SJLS	Dyk1252+411	1	76 / 17

DyK1017+205 with a three-step prediction horizon, and SJLS Dyk1252+411 with an external TSP-derived velocity field. The learned edge and surface relevance values are response-consistent model diagnostics; they are not direct measurements of contact force, contact pressure, jamming probability, or engineering risk.

3.5 Experimental setup

All formal runs use a mileage-ordered split so that testing is performed on a later continuous tunnel interval. The BSLL one-step run contains 30 training samples and 8 test samples (target chainages 41–48 m). The BSLL three-step run contains 29 training samples and 7 test samples (target chainages 42–48 m). The SJLS run contains 76 training samples and 17 test samples (target chainages 99–115 m). The graph construction uses $\tau_{\text{edge}} = 2.0$ m, $\eta_{\text{min}} = 0.3$, and $\tau_{\text{zone}} = 5.0$ m. The temporal history window is $K = 5$.

3.6 Forward prediction across the two tunnel cases

Table 2 reports the case-level forward prediction metrics. The results support a cautious interpretation. In BSLL, the one-step graph-sequence model obtains MAE 0.415, essentially matching Persistence (0.415) and the random-edge variant (0.415). The three-step BSLL run similarly remains close to Persistence (0.462 versus 0.462), while the random-edge variant is slightly lower in MAE (0.461). These BSLL results do not support a claim of predictive superiority.

The SJLS case gives a more favourable but still small effect. The full model obtains MAE 0.2553, slightly lower than Persistence (0.2554), randomised edges (0.2559), and the no-prior variant (0.2570). The Pearson correlation is high ($r = 0.960$), but R^2 remains negative because the mileage-ordered test interval is a demanding extrapolation setting. Thus the SJLS result is best read as a small response-consistency advantage under the external TSP case, not as a large accuracy breakthrough.

Table 2: Forward prediction results for the semantic case runs. MAE and RMSE are computed on standardised multivariate response targets. The random-edge column reports the corresponding graph-topology ablation.

Case	Test n	Full MAE	Full RMSE	R^2	Persistence MAE	Random-edge MAE
BSLL h=1	8	0.415	0.593	−0.621	0.415	0.415
BSLL h=3	7	0.462	0.554	−0.631	0.462	0.461
SJLS h=1	17	0.255	0.347	−0.482	0.255	0.256

Figure 3 illustrates the prediction comparison for the representative BSLL one-step run. The figure should be interpreted as a diagnostic example rather than as the central evidence for the paper.

3.7 Graph ablations and the role of geometry

The ablations separate two questions that should not be conflated: whether the graph-sequence representation improves prediction, and whether geometry-constrained relations produce interpretable spatial diagnostics. Prediction evidence for the first question is mixed. In SJLS, removing the geometric prior increases MAE from 0.2553 to 0.2570, and randomising rock-machine edges increases MAE to 0.2559. This supports a small geometry-sensitive effect in the external TSP case. In BSLL, however, randomised edges are comparable to or slightly better than the full model. The BSLL ablation therefore should not be used to claim that constrained topology improves forecasting accuracy.

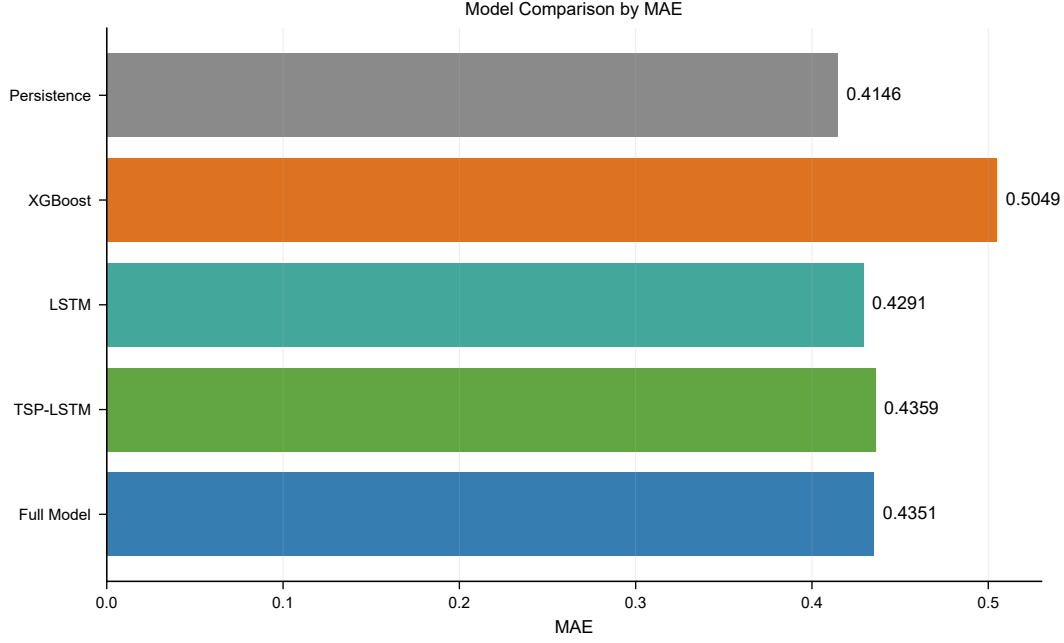


Figure 3: Prediction comparison for the BSLL one-step run on the mileage-ordered future interval. The graph-sequence framework is comparable to temporal baselines but does not establish a predictive advantage in this case.

Table 3: Geometry-related ablation summary. The SJLS case shows a small advantage for the full geometry-aware model, whereas BSLL mainly supports the availability of graph-based interpretation rather than a robust accuracy gain.

Case	Full MAE	Random-edge MAE	No-prior MAE
BSLL h=1	0.415	0.415	0.415
BSLL h=3	0.462	0.461	0.461
SJLS h=1	0.255	0.256	0.257

Figure 4 shows the BSLL ablation pattern. Its main value is negative evidence: it prevents overclaiming and clarifies that the paper’s stronger contribution lies in spatially explicit representation and interpretation.

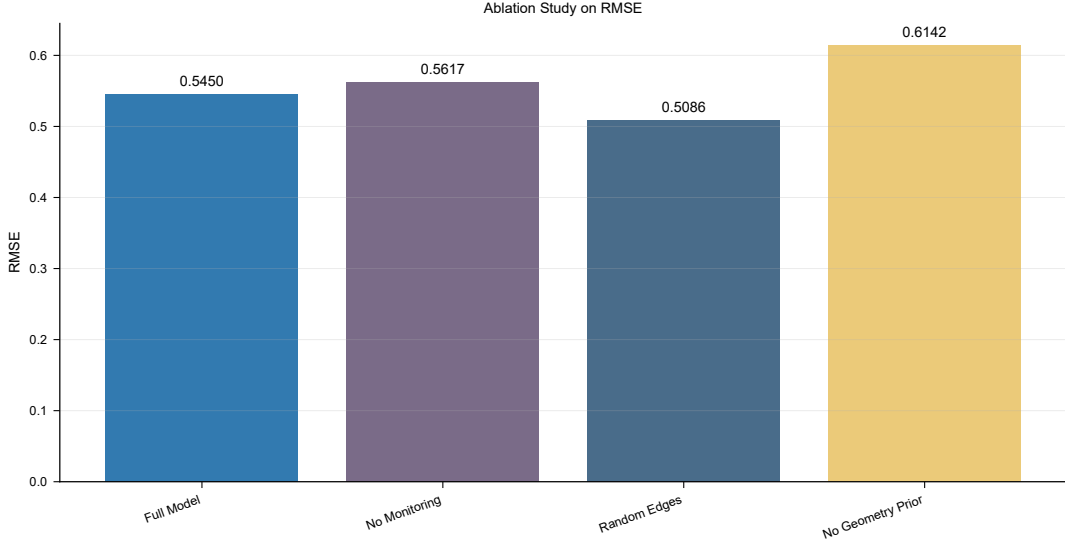


Figure 4: Ablation comparison for the BSLL one-step run. The random-edge variant is not worse in this small shifted test interval, so the BSLL result does not support a strong topology-accuracy claim.

3.8 Spatial consistency of learned surface relevance

The interpretation evidence is more consistent across cases. Table 4 reports four post-hoc indicators computed from the saved checkpoints. Surface relevance is positively spatially auto-correlated in all three runs: Moran’s $I = 0.546$ for BSLL $h=1$, 0.302 for BSLL $h=3$, and 0.300 for SJLS. This indicates spatially organised relevance fields rather than random surface noise. The degree-control correlations are negative in all cases (-0.229 , -0.324 , and -0.456), which argues against a trivial explanation in which nodes appear important only because they have more incident candidate edges.

The geology association differs by case. BSLL shows a strong negative association between mean surface relevance and local TSP velocity ($r = -0.989$ for $h=1$ and $r = -0.688$ for $h=3$), whereas SJLS shows a strong positive association ($r = 0.870$). The signs should not be interpreted as universal physical rules; they indicate that learned relevance co-varies with the local TSP field within each tunnel setting. The strongest response-linked association also differs: AdvanceRate for BSLL and Torque for SJLS. This case-specific behaviour shows that the graph-to-surface route can expose different response-relevance patterns rather than producing a single generic hotspot template.

3.9 Spatial interpretation outputs

Figures 5 and 6 show representative BSLL interpretation outputs. The hotspot map projects response-supervised surface relevance onto the TBM geometry, while the chainage view tracks mean surface relevance together with monitoring responses over the future interval. These outputs are the main added information supplied by the graph-sequence representation: monitoring-only baselines can predict response variables but cannot map model relevance back to TBM surface components or rock-machine candidate relations.

Table 4: Post-hoc spatial interpretation evidence. Rel.-Geo. r is the Pearson correlation between mean surface relevance and TSP V_p across test chainages. Degree r correlates surface relevance with incident candidate-edge degree; negative values indicate that relevance is not a direct positive proxy for edge count.

Case	Moran's I	Geometry-only I	CV	Degree r	Rel.-Geo. r
BSLL h=1	0.546	-0.054	0.018	-0.229	-0.989
BSLL h=3	0.302	-0.054	0.005	-0.324	-0.688
SJLS h=1	0.300	0.196	0.014	-0.456	0.870

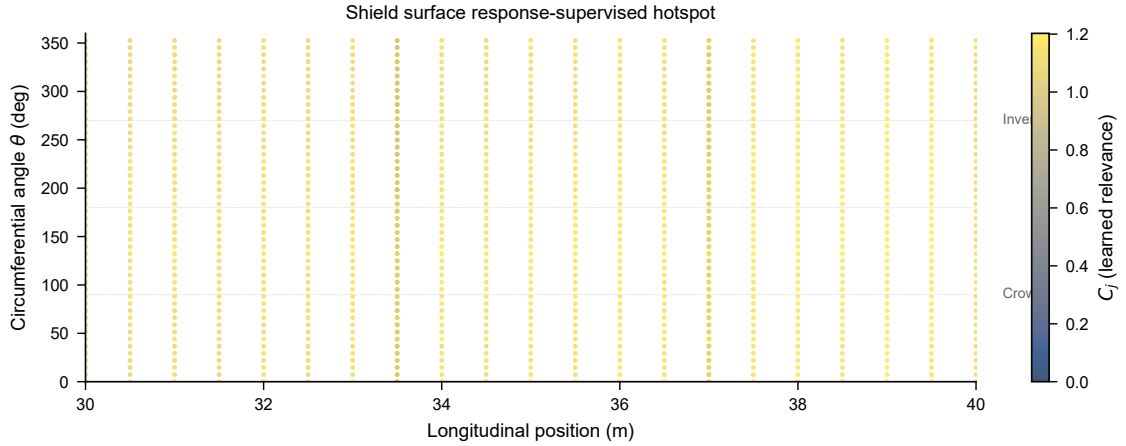


Figure 5: Representative response-consistent TBM surface relevance map for the BSLL one-step run. Colour intensity indicates degree-normalised surface relevance and is used as a diagnostic visualisation, not as a measured contact pressure field.

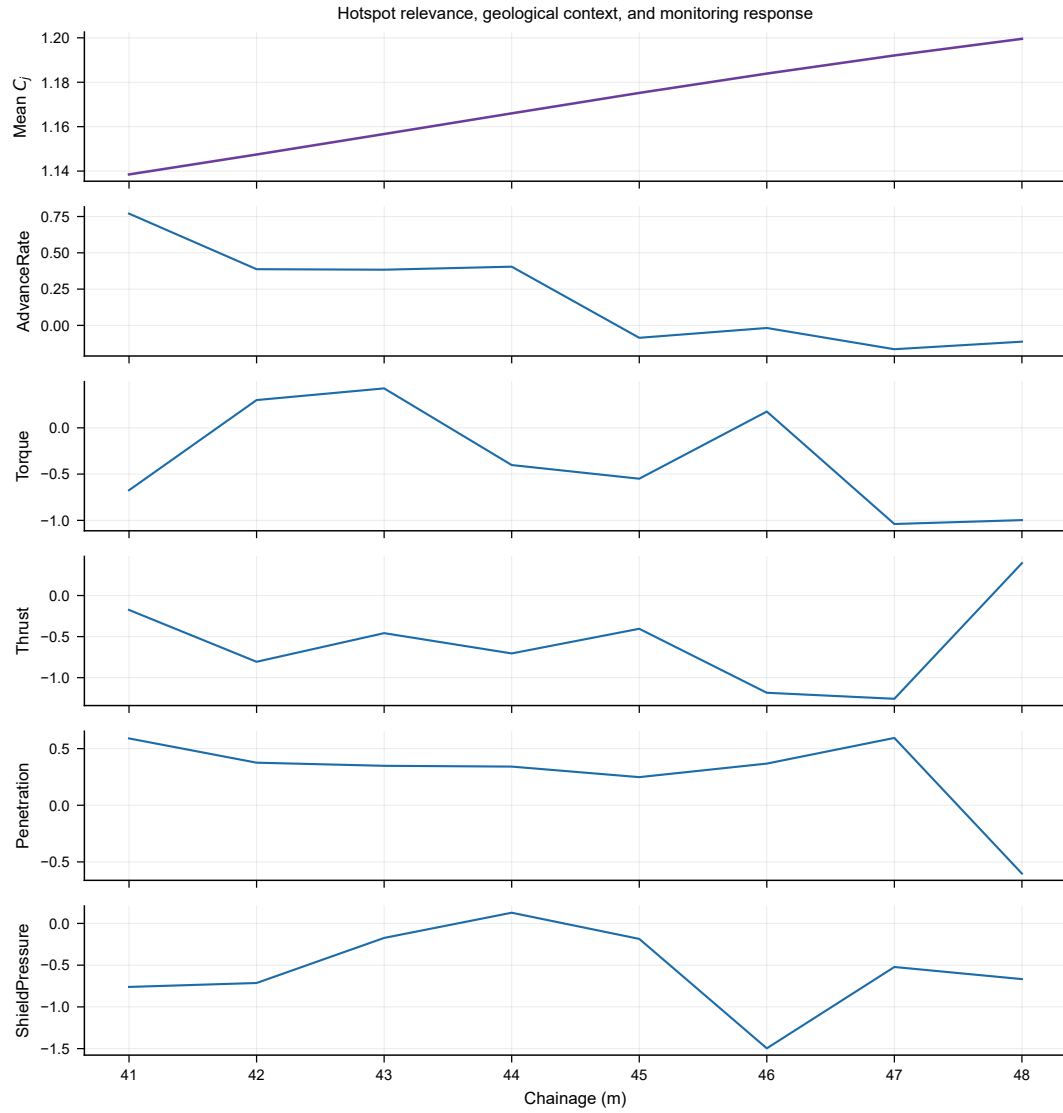


Figure 6: Representative chainage evolution of surface relevance and monitoring responses for the BSLL one-step run. The view illustrates how the learned diagnostic can be traced along excavation progress.

3.10 Robustness and remaining checks

The current evidence supports three bounded claims: the framework remains competitive with temporal baselines, SJLS shows a small geometry-sensitive accuracy effect, and all three saved checkpoints produce spatially organised surface relevance. It does not support claims of statistically significant global accuracy improvement, measured physical interaction recovery, or operational risk calibration. Multi-seed stability, label permutation, threshold sensitivity, spatial detrending, and additional cross-project validation remain necessary before stronger robustness claims can be made.

4 Discussion

4.1 What the current experiments actually support

The revised experiments sharpen the paper’s central claim. The framework should not be presented as a broadly superior forecasting model. In BSLL, the one-step and three-step runs are essentially tied with Persistence and do not benefit clearly from constrained topology. In SJLS, the full model gives a small MAE advantage over Persistence, randomised edges, and the no-prior variant, but the gain is modest. This pattern is meaningful because it separates accuracy from representation: the model is not winning by a large margin, but it provides a spatial entity-relation structure that temporal baselines cannot produce.

This distinction changes the interpretation of the contribution. The prediction task is used as response supervision and as a consistency check. It is not the only, or even the strongest, evidence for the method. The stronger evidence is that TSP voxels, TBM surface components, candidate rock-machine relations, monitoring responses, and projected relevance diagnostics remain connected in a single spatial representation. The paper’s value therefore lies in making rock-TBM interaction inspectable as a spatial process under incomplete observability.

4.2 Why the two tunnel cases are complementary

BSLL and SJLS should be read as complementary evidence rather than as a development-validation pair. BSLL provides a compact case in which the limits of prediction are visible: the model is competitive but does not outperform simple temporal persistence, and randomised edges are not worse. This is useful because it prevents the paper from overclaiming geometry as an accuracy mechanism. At the same time, BSLL provides interpretable multi-step and surface-evolution examples that illustrate what a graph-sequence representation adds beyond scalar prediction.

SJLS plays a different role. It uses an external TSP-derived velocity field and a larger number of forward test samples. In this case, the full geometry-aware model is slightly better than the ablated variants, and the learned relevance is strongly associated with the local TSP field. This makes SJLS the better case for arguing that geometry-constrained representation can support response-consistent prediction and spatial interpretation. The important point is not that SJLS proves a large accuracy gain, but that it shows the same representation operating in a second tunnel setting with a clearer geometry-sensitive signal.

4.3 Spatial interpretation as the main added value

The most consistent result across the experiments is spatial organisation of the learned surface relevance. Moran’s I is positive in all three runs, and the degree-control correlations are negative. These two facts support a bounded but useful claim: the surface relevance maps are not spatially random and are not simply inflated by the number of candidate edges incident to a TBM node.

This does not make the maps physical contact-force maps. The relevance values are produced by a response-supervised neural model and remain conditional on the forecasting objective, graph construction rules, and training data. Their proper role is diagnostic: they indicate where the model places response-linked importance on the TBM surface under geometry-screened rock-machine candidate relations. This diagnostic role is still valuable because conventional monitoring-only sequence models cannot localise response relevance on a TBM surface or compare cutterhead and shield regions.

4.4 Implications for GIScience representation

The experiments motivate a GIScience framing of TBM excavation analysis. The problem is not merely to add more features to a forecasting model. It is to formalise heterogeneous spatial entities with different supports: TSP voxels, TBM surface nodes, rock-machine candidate relations, and chainage-indexed monitoring responses. Once these entities are represented explicitly, model outputs can be projected back into spatial views that support inspection, comparison, and case-based reasoning.

This representation perspective also explains why prediction and spatial diagnosis must be evaluated separately. A temporal baseline may be difficult to beat on short, autocorrelated monitoring series, especially under a mileage-ordered split. However, such a baseline cannot answer where the model places interaction relevance, whether relevance is spatially clustered, or how the relevance pattern differs between BSLL and SJLS. The graph-sequence framework contributes this second evaluation axis.

4.5 Limitations and next steps

Several limitations remain. The test sets are small, especially for BSLL, and all results are based on single fixed-seed formal runs. Negative R^2 values show that forward extrapolation remains difficult under geological shift. The observed attention-geology correlations may partly reflect chainage trends and therefore require spatial detrending before stronger geological interpretation is made.

Future work should prioritise multi-seed stability, label-shuffle controls, candidate-edge threshold sensitivity, spatial detrending, and additional tunnel sections. Independent event labels, expert assessments, or physical measurements would be required before the relevance maps could be connected to operational risk or actual contact conditions. Until then, the defensible claim is representation and response-consistent interpretation, not physical recovery or general predictive superiority.

5 Conclusion

This study proposed a response-supervised, geometry-constrained graph sequence framework for spatially explicit rock-TBM interaction representation. The method formalises TSP-derived rock voxels and TBM surface components as heterogeneous spatial entities, screens candidate rock-machine relations by geometric compatibility, and projects learned relevance back to TBM surface and chainage views.

The revised experiments clarify the strength and boundary of the contribution. Across BSLL DyK1017+205 and SJLS Dyk1252+411, the framework remains competitive with temporal baselines but does not establish a broad, statistically significant accuracy advantage. BSLL mainly supports the multi-step and surface-interpretation use case, while SJLS provides the clearest small geometry-sensitive prediction effect. The consistent evidence is interpretive: all three saved checkpoints produce positively clustered surface relevance, and the degree-control checks indicate that relevance is not simply a proxy for candidate-edge count.

The main contribution is therefore not a claim of large forecasting superiority but a spatial entity-relation representation and an auditable route from response supervision to surface-based diagnostics. Future work should test multi-seed stability, label-shuffle controls, threshold sensitivity, spatially detrended geology association, and additional tunnel cases before stronger claims are made about physical interaction recovery or operational risk estimation.

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Data availability statement

State where the data, code, and materials can be accessed, and note any restrictions if applicable.

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